

# Flexible Cable Simulation in NVIDIA Isaac Sim

Comparing three modelling approaches for RL-based cable manipulation

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- **Why cables?** wire-harness assembly, cable routing, and deformable-object manipulation are common, unsolved robotics tasks.
- **The challenge:** Isaac Sim has *no native 1-D deformable (rod/cable) body* – only rigid bodies and volumetric FEM soft bodies.
- **This study:** implement three candidate cable models and lay out *what each can and cannot do* – their capabilities and limitations – as a basis for choosing one later for cable manipulation.

Three approaches, each in two scenarios (free cable, and cable held by a Franka arm). This talk presents trade-offs and limitations – it does not pick a winner.

# Three approaches

## 1. Capsule chain

rigid capsules + D6 joints; bending from Euler–Bernoulli beam theory (PhysX).

## 2. Cosserat rod (Warp)

1-D position-based elastic rod in NVIDIA Warp, after the *OGC SIGGRAPH-2025* paper.

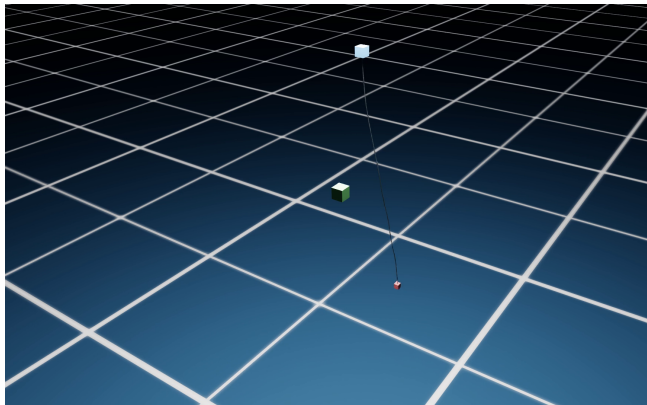
## 3. FEM deformable

Isaac Sim's built-in volumetric soft body (tetrahedral FEM).

Two scenarios each  $\Rightarrow$  **6 simulations**: (A) free hanging/swinging cable manipulator. (B) cable connected to a robot

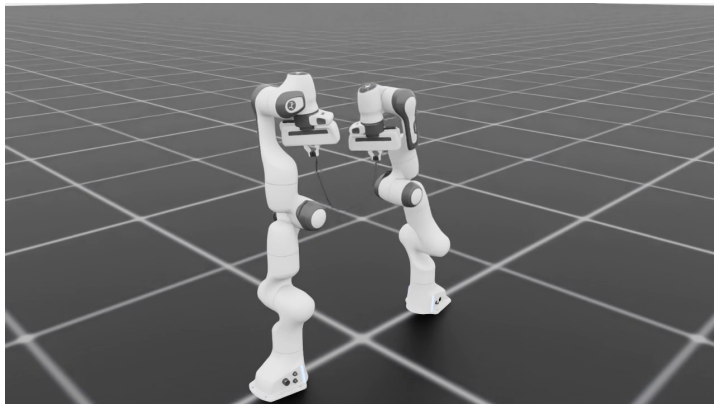
# Approach 1 — Capsule chain (D6 joints)

- Rigid capsules linked by **D6 joints**; translations locked (inextensible), bending springs  $k = EI/L$  from beam theory.
- PhysX **TGS** solver, up to 200 links, 240 Hz.
- Reproduces the **Govoni et al. 2025** stability sweep.
- Modes: hanging-kick and both-ends-fixed.



Video: *approach1\_cable\_only/video.mp4* (shared folder)

# Approach 1 — With robot

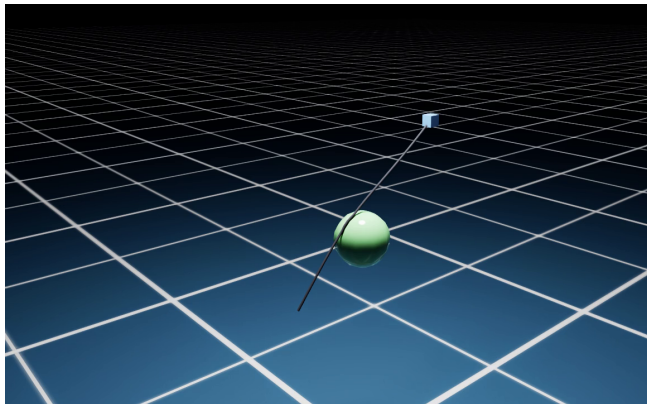


*Video: approach1\_robot/video.mp4 (shared folder)*

Inextensible (rigid joints): a hard pull **drags the second arm**  $\sim 17$  cm while the cable barely stretches ( $\sim 5$  cm)  
 $\Rightarrow$  **true force transmission.**

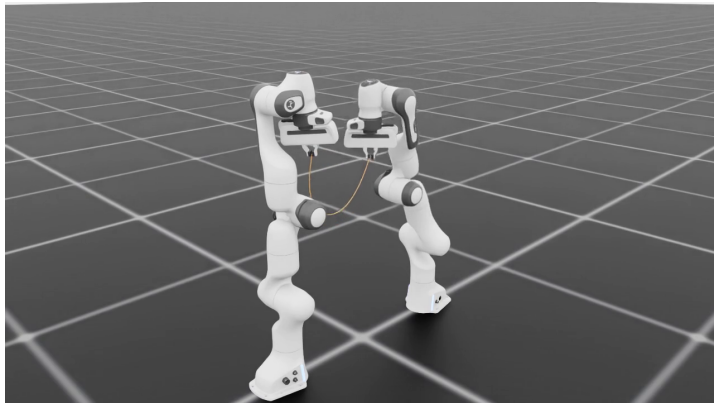
## Approach 2 — Cosserat rod in NVIDIA Warp

- 1-D **Cosserat** / **XPBD** elastic rod: inextensible stretch + bending stiffness + gravity, solved on the GPU in **Warp**.
- Idea from the **OGC** paper (Chen et al., SIGGRAPH 2025): penetration-free contact for cloth & rods.
- Truly **thin** (no FEM fattening); **exact** node-vs-primitive collision.



Video: *approach2\_cable\_only/video.mp4* (shared folder)

## Approach 2 — With robot

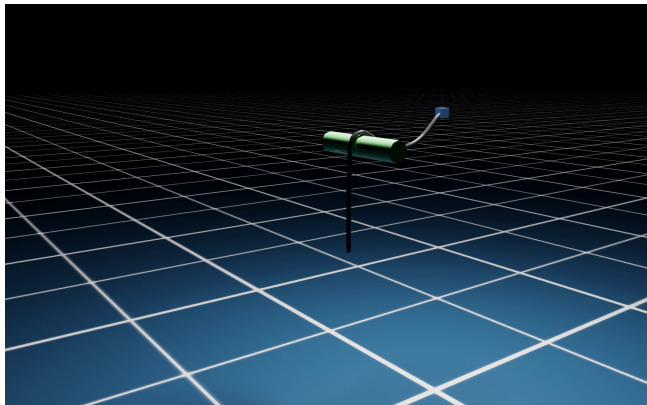


*Video: `approach2_robot/video.mp4` (shared folder)*

End nodes **pinned** to the grippers: the thin rod follows the arms and sags cleanly between them (best-looking cable). **One-way** – it follows but exerts no force back, so it doesn't drag the second arm.

## Approach 3 — FEM deformable (Isaac Sim)

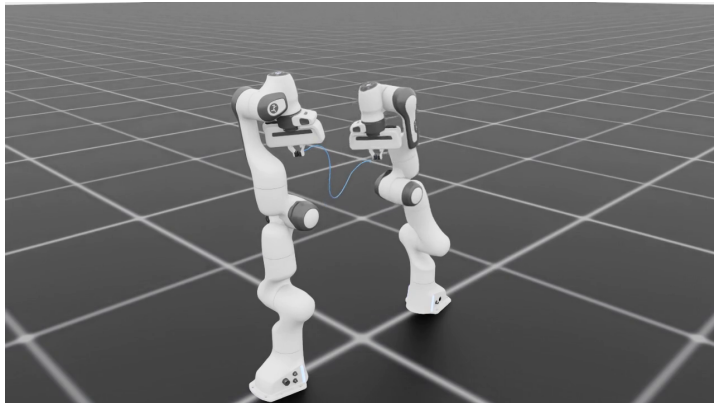
- Isaac Sim's built-in **volumetric** (tetrahedral) FEM soft body.
- TPU material:  $E = 40 \text{ MPa}$ ,  $\nu = 0.48$ ,  $\rho = 1150 \text{ kg/m}^3$ .
- Rod **fattened** to 12 mm and  $E$  rescaled so it bends/weights like the real 1.5 mm cable (voxeliser needs  $\geq 3$  hexes across).
- Rendered thin via a swept tube on its centreline.



Video: *approach3\_cable\_only/video.mp4* (shared folder)



## Approach 3 — With robot



*Video: approach3\_robot/video.mp4 (shared folder)*

Region-weld attachment. Being a soft body it **stretches a lot** under a hard pull ( $\sim 20\text{--}35\%$ ) and drags the follower little  $\Rightarrow$  poor force transmission.

# Trade-offs at a glance

| Criteria             | D6 capsule       | Cosserat (Warp)                     | FEM                               |
|----------------------|------------------|-------------------------------------|-----------------------------------|
| Cable realism        | beam-bending     | 1-D rod (thinnest)                  | volumetric (jelly/stretch)        |
| Stretches?           | no (rigid links) | no (inextensible)                   | yes ( $\sim 10\text{--}50\%$ )    |
| Thinness             | true 1.5 mm      | true (any radius)                   | $\geq 12$ mm (faked thin)         |
| Compute cost         | low              | low ( $\sim 1.4 \times \text{RT}$ ) | high ( $< 0.2 \times \text{RT}$ ) |
| Robot force coupling | two-way          | one-way (or spring)                 | two-way (rough)                   |
| Collision            | exact capsules   | exact (coded prims)                 | approximate tets                  |
| Setup / code         | many joints      | custom Warp solver                  | built-in                          |

RT = real time. These are *trade-offs* – no single method wins on every row; the right one depends on the priority.

# Pros and cons (each is a trade-off — no single “best”)

## 1. Capsule chain (D6 joints)

- + Stable; physically grounded; real-time; two-way force coupling; easy to randomize.
- Piecewise-rigid (no axial give); many bodies; needs a heavy solver to stay inextensible.

## 2. Cosserat rod in Warp (OGC)

- + Thinnest, cleanest cable; inextensible; no jelly; exact node-vs-primitive collision; GPU-fast.
- Separate (non-PhysX) sim; only artificial robot coupling; hand-coded collision; custom solver.

## 3. FEM deformable

- + True volumetric soft body; native to Isaac Sim; two-way; handles squashing/draping.
- Fat, stretches, jelly, rigid at the joint, rough collision, slow.

# Limitations — Capsule chain (D6 joints)

- **Piecewise-rigid:** links cannot stretch *at all* – it is a chain of rigid bodies, not a continuum; bending is a discrete per-joint spring.
- **Many bodies:** needs 60–200 links + joints for a smooth look  $\Rightarrow$  many rigid bodies and constraints to simulate.
- **Solver cost vs. rigidity:** staying inextensible under load needs high position-iteration counts ( $\approx 180$ ); too few  $\Rightarrow$  it stretches, too many  $\Rightarrow$  it goes unstable.
- **Numerical twist:** the free twist DOF spins up unless explicitly damped.
- **Tuning-sensitive:** link mass / joint stiffness / damping / timestep must be balanced (mass-ratio & dt limits) or the chain diverges.
- **Approximate bending:** torsional springs from beam theory, not a true rod.

# Limitations — Cosserat rod (Warp / OGC)

- **Not a native physics body:** it is a *separate* Warp simulation  $\Rightarrow$  it does not interact with PhysX bodies through the engine (no automatic contact with robots or objects).
- **Artificial robot coupling:** either kinematic (**one-way** – follows the gripper, no force back) or a hand-tuned spring (two-way, but then it visibly *stretches*); no clean physical two-way coupling.
- **Hand-coded collision only:** node-vs-primitive (ground, sphere) that you write; no general mesh contact, no self-collision unless coded.
- **Implementation burden:** you must write & maintain the XPBD solver kernels – not off-the-shelf; every new obstacle needs custom contact code.

# Limitations — FEM deformable

- **Cannot be thin:** voxeliser needs  $\geq 3$  hexes across the diameter and cooking caps resolution  $\Rightarrow$  thinnest  $\approx 12$  mm (real cable 1.5 mm,  $8\times$  too thick); only *rendered* thin.
- **Stretches:** one isotropic  $E$  ties bending ( $\propto Er^4$ ) to axial ( $\propto Er^2$ ) – exact bending  $\Rightarrow$  axially soft ( $\sim 50\%$  stretch); firm enough not to stretch  $\Rightarrow$  bending  $20\text{--}30\times$  too stiff.
- **“Jelly”:** low-resolution soft body + near-incompressible  $\nu \Rightarrow$  volumetric locking / wobble.
- **Rigid at the connection:** attachment *welds a region* of vertices (clamped-beam boundary)  $\Rightarrow$  straight/stiff near the joint, not a pin.
- **Rough collision:** simplified, re-meshed coarse collision tets  $\Rightarrow$  imprecise, can clip thin obstacles; self-collision off by default.
- **Expensive:** GPU FEM runs  $< 0.2\times$  real-time.

Full derivations: `why_fem_cant_be_thin.py`, `README_fem_attachment_collision.md`.

Thank you

Questions?

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